THE NOVA RESEARCH CURRICULUM

Chemistry

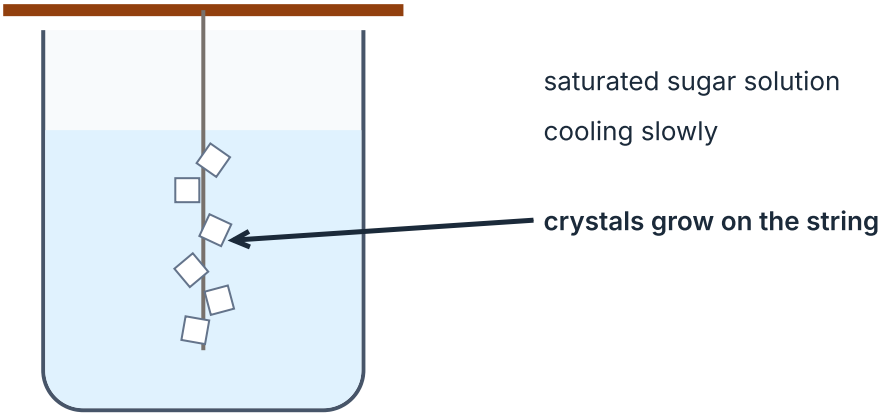
Grade 7 · Term 3

Student edition

Outcome: CH-7.3 solutions, solubility and saturation; factors affecting solubility.

Lesson notes

- **Solutions (Weeks 1–4):** solute + solvent → solution (sugar + water). Water dissolves many substances. Soluble and insoluble substances (salt vs sand). Concentrated and dilute solutions.
- **Solubility practical (Weeks 5–8):** fair test: how does water temperature affect how much sugar dissolves? Change one variable (temperature); keep the others the same (amount of water, stirring); measure spoonfuls dissolved. Full practical report: question, prediction, variables, method, results table, line graph, conclusion. Hot water only with an adult.
- **Saturation (Weeks 9–12):** a saturated solution can dissolve no more solute at that temperature. For most solids, solubility rises with temperature; gases dissolve less in warm water (a fizzy drink goes flat faster when warm). Crystals grow from a cooling saturated solution.



Crystals grow from a cooling saturated solution.

Workbook pages

Page	Task
1	Label solute, solvent and solution for six examples.
2–3	Solubility practical report with a line graph.
4	Crystal-growing log.

Question bank

#	Outcome	Question	Your answer
1	CH-7.3	In salt water, salt is the …	solvent / solute / solution / precipitate
2	CH-7.3	Which is insoluble in water?	sugar / salt / sand / coffee powder
3	CH-7.3	A solution that can dissolve no more solute is …	dilute / saturated / insoluble / empty
4	CH-7.3	For most solids, solubility increases as temperature …	falls / rises / stays the same / disappears
5	CH-7.3	Fizzy drinks go flat faster when warm because gases dissolve less in warm water. True or false.	True / False
6	CH-7.3	In a fair test, how many variables do you change?	one / two / three / all of them
7	CH-7.3	Name two variables to keep the same in the sugar test.	_____